

KOUSHIK RAM

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PROFESSIONAL SUMMARY

Software Engineer with **1+** year of experience delivering production-grade desktop and cloud-native applications across aerospace and enterprise domains shipping real hardware-integrated systems for **HAL's ALH helicopter program**, serverless platforms on **AWS**, and **full-stack web applications** using **React.js, Node.js, and Python**. Proven ability to own features end-to-end, from NoSQL schema design and RESTful API development to CI/CD pipelines, multi-threaded GUI architecture, and hardware communication layers over **VISA/SCPI**. **Strong foundation in OOP, MVC, Agile delivery, and secure coding** with a track record of reducing defects by **30%**, cutting latency by **40%**, and shipping zero-defect handoffs to defense clients.

SKILLS

Languages: C, C++, Python, JavaScript, HTML, CSS

Back-End & DBs: Node.js, Java (Spring Boot basics), Python, MySQL, PostgreSQL, MongoDB, SQLite, DynamoDB, AWS S3

Testing: Selenium, POM, JMeter, Jest, Mocha, Jasmine, Postman, System & Integration Testing, REST APIs, Test plan, Bug tracking, Regression Testing

Front-End: React.js, Vue.js, Next.js, Tailwind CSS, Responsive UI/UX

DevOps & Tools: Docker, Jenkins, Kubernetes, GitHub Actions, Git, VS Code, Visual Studio

Practices & OS: DSA, OOP, Domain-Driven Design, OS (Windows 10/11, Linux), DBMS, Agile, CI/CD, Secure Coding, Scalability

EXPERIENCE

Software Development Engineer — GUI & Software Development | Zing Technologies

Aug 2025 - Present

PyQt5 • Python • MongoDB • SQLite • MVC Architecture • Aerospace Domain

- **Architected** a production-grade, multi-threaded PyQt5 desktop application for **HAL's NextGen Airborne Intercom System (ISTJ)** by applying **MVC architecture** and **system design principles**, meeting stringent aerospace reliability requirements.
- **Delivered** a serverless task management platform (**Python, MongoDB, AWS Lambda**) as sole contributor by designing the **NoSQL data model** and RESTful Lambda endpoints, cutting manual follow-up overhead by **80%** and driving adoption across 7 teams.
- **Streamlined** full-stack integration across 5+ application modules by connecting React UIs to AWS Lambda and third-party REST services using an API-first approach, reducing cross-module defects by **30%**.
- **Accelerated** test execution by **40%** by diagnosing and resolving **SQLite** performance bottlenecks via query analysis while enforcing **OOP** and secure coding standards across code reviews.
- **Validated** core workflows across the full SDLC by writing unit and integration tests and independently tracing defects across a multi-module codebase.

Software Developer Trainee — Automation & CI/CD | Pyspiders

Jun 2025 – Aug 2025

Python • Jenkins • Docker • Kubernetes • GitHub Actions

- **Designed** two Python microservices automation frameworks (**ShopperStack, QBank**) with independently deployable, loosely coupled modules covering API design, integration testing, and **CI/CD** deployment.
- **Established** end-to-end CI/CD pipelines using **Jenkins, Docker, Kubernetes, and GitHub Actions** by standardizing build and release workflows, reducing release cycle time and improving cross-environment reliability.
- **Traced** and resolved critical system-layer **defects** in a commercial multi-module codebase by independently performing **root cause analysis** with minimal supervision.

PROJECTS

ISTJ Intercom System Test Jig (Defense Grad GUI) | [GitHub](#)

Dec 2025 – Ongoing

- **Spearheaded** a real-time, multi-threaded **PyQt5** desktop application for **HAL's NextGen Airborne ALH helicopter** system as **sole contributor** by applying concurrent data stream handling and state management best practices from ground up.
- **Integrated** a multi-instrument hardware communication layer across **4+** bench instruments over **VISA/SCPI** for **station box and junction box** testing by implementing deterministic test sequencing across 5 connector groups, **reducing manual intervention by 60%**.
- **Fortified** system reliability across **PSU, audio analyzer, DMM, oscilloscope, and ISTJ board** by implementing active listening, self-test sequencing, single-test tuning, and a 3-tier admin approval workflow, cutting mid-test abort rates by **40%** and ensuring **100% repeatable results**.
- **Structured** data logging and query-based analysis by applying MVC architecture with an SQLite backend, enabling robust integration testing across the full test suite.

Task Management System — Internal Productivity Tool | GitHub

Aug 2025 – Dec 2025

- **Shipped** an internal task management desktop app (PyQt5, MongoDB) by building **sprint planning, task logging, and event scheduling modules** with manager and company-head dashboards, driving team-wide visibility across 7 teams.
- **Incorporated AWS Lambda and DynamoDB** as a serverless backend by designing the NoSQL schema, implementing CRUD operations, and writing Jest unit tests, enabling real-time task sync and automated reminders org-wide.
- **Packaged** the application as a standalone Windows installer via **Inno Setup** by configuring one-click deployment, eliminating environment setup for non-technical end users.

- **Optimized** a *full-stack music streaming app* (React.js, Next.js, Node.js) by applying lazy loading, memoization, and efficient state management, improving load time by **40%** with cross-browser responsive UI.
- **Achieved 85%** Jest test coverage across **12+ React components** by implementing component-level unit tests and integrating test runs into the GitHub Actions CI pipeline to catch regressions before every merge.

EDUCATION

Bachelor of Information Science & Engineering

MVJ College of Engineering, Bengaluru

CGPA: 8.538

Jul 2025

ACHIEVEMENTS

- **Ranked** globally at **3392** on LeetCode by consistently solving advanced DSA problems across data structures and algorithms.
- **Handed** off the ISTJ software to HAL with all client requirements met and zero critical defects by maintaining end-to-end ownership and delivery discipline throughout the SDLC.

EXTRACURRICULAR ACTIVITIES

- **Mobilised** a cross-functional team as **Event Coordinator at SDC, MVJCE** by *conceptualizing and executing hackathons and technical events, representing the club at inter-college competitions.*
- **Demonstrated** leadership as **House Captain at Air Force School ASTE** by *coordinating large-scale events and driving team cohesion, strengthening communication and organizational discipline.*

CERTIFICATIONS

- Career Essentials in GitHub – *Github*
- Databricks Fundamentals – *databricks*
- Azure AI Essentials – *Microsoft*